

Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 data.

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ASTRONOMY & ASTROPHYSICS: HELIOPHYSICS WITH ADITYA L1 DATA

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PART 2A

Tracking the Evolution of Solar Active Regions

Abstract

The objective of this study was to track the temporal evolution of solar active regions using SUI observations from Aditya-L1. Multiple-day observations were intended for analysis. However, due to the limited availability of downloadable observations through the PRADAN archive during the project period, a complete temporal evolution study could not be performed. The available SUI observation was processed to demonstrate the workflow for active region detection.

Introduction

Solar active regions evolve continuously and are associated with strong magnetic activity. Monitoring their evolution helps understand solar dynamics and space weather. In this work, the planned approach was to analyze a sequence of SUI observations obtained over several days.

Methodology

- Download SUI FITS observations.
- Preprocess the solar image.
- Detect active regions.
- Measure the active-region area.
- Compare observations across multiple days.

Dataset

Instrument: Aditya-L1 SUI

Data Type: FITS Image

Observation: Solar Ultraviolet Image

Results

The available SUI observation was successfully processed. The active region was identified using image segmentation techniques. Due to the limited availability of

different daily observations, temporal comparison across multiple days could not be completed.

Discussion

The workflow for tracking solar active regions was implemented successfully. The downloaded SUI observation demonstrated the image processing pipeline. However, the PRADAN archive returned the same observation for different search dates, preventing quantitative temporal analysis.

Conclusion

The methodology for monitoring solar active regions was successfully implemented. Future work may include additional SUI observations collected over multiple days to quantify changes in active-region area over time.

PART 2B

Intensity–Magnetic Flux Correlation using Aditya-L1 SUIT and MAG Data

Abstract

Solar active regions are characterized by enhanced brightness and strong magnetic activity. Studying the relationship between solar intensity and magnetic field helps in understanding solar dynamics and space weather phenomena. In this work, SUIT (Solar Ultraviolet Imaging Telescope) images from Aditya-L1 were processed to detect active regions using image segmentation techniques. The detected regions were compared with magnetic field measurements obtained from the Aditya-L1 MAG instrument. The average intensity of the active region and the corresponding magnetic field magnitude were calculated. Since the available MAG dataset contains spacecraft magnetic field measurements rather than image-based magnetograms, a direct pixel-wise correlation could not be performed. Nevertheless, the study demonstrates a workflow for integrating solar imaging and magnetic field observations.

Keywords: Aditya-L1, SUIT, MAG, Active Region, Image Segmentation, Magnetic Field, Solar Imaging.

1. Introduction

Solar active regions are localized areas on the Sun exhibiting enhanced magnetic activity and increased ultraviolet emission. These regions are often associated with solar flares, coronal mass ejections, and other energetic events that influence space weather.

The Solar Ultraviolet Imaging Telescope (SUIT) aboard Aditya-L1 provides ultraviolet images of the solar disk, enabling the identification of active regions. The Aditya-L1 Magnetometer (MAG) measures the magnetic field around the spacecraft.

The objective of this study is to investigate whether a measurable relationship exists between the brightness intensity of active regions detected in SUIT images and the magnetic field measurements provided by the MAG instrument.

2. Objectives

The objectives of this work are:

- Preprocess the SUIF FITS image.
- Detect active regions using image segmentation.
- Calculate the mean brightness intensity of the detected region.
- Extract magnetic field measurements from the MAG dataset.
- Compare image intensity with magnetic field magnitude.
- Evaluate the possibility of intensity–magnetic field correlation.

3. Dataset Description

Two datasets were used.

3.1 SUIF FITS Image

The solar image was obtained from the Aditya-L1 SUIF instrument.

The image contains ultraviolet observations of the solar disk showing brightness variations corresponding to active regions.

Dataset Information

Parameter	Description
Instrument	SUIF
Data Type	FITS Image
Observation	Solar UV Image
Purpose	Active Region Detection

3.2 MAG Dataset

The magnetic field measurements were obtained from the Aditya-L1 MAG instrument.

The NetCDF (.nc) file contains:

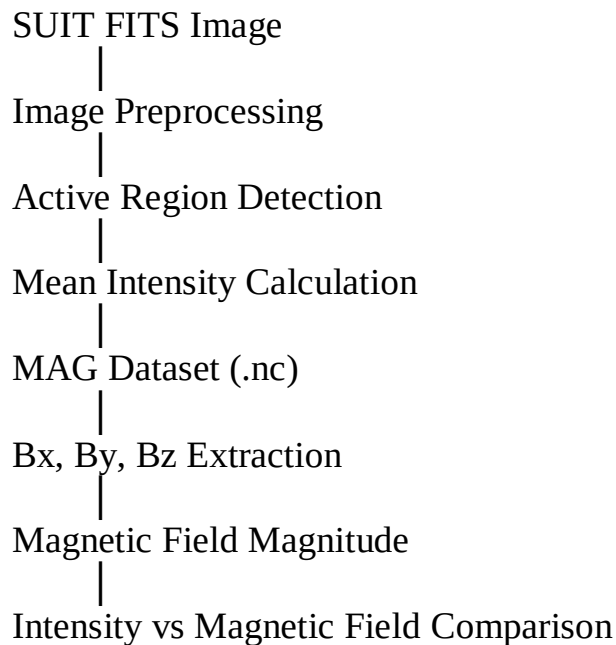
- B_x (GSE)
- B_y (GSE)

- Bz (GSE)
- Bx (GSM)
- By (GSM)
- Bz (GSM)

These represent spacecraft magnetic field measurements.

4. Methodology

The overall workflow is shown below.



5. Image Preprocessing

The SUIT FITS image was preprocessed before segmentation.

The preprocessing steps included:

- Reading the FITS image.
- Removing invalid values.
- Image normalization.
- Contrast enhancement.
- Noise reduction.

These operations improve the visibility of solar features and enhance segmentation accuracy.

6. Active Region Detection

The processed image was segmented using Otsu thresholding.

The segmentation separates brighter active regions from the background solar disk

Small isolated noise components were removed using morphological filtering.

Detected Active Region

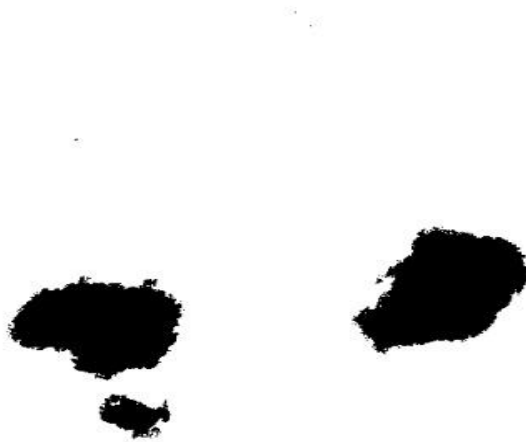


Figure 1. Detected Active Region

7. Magnetic Field Extraction

The NetCDF magnetic field dataset was read using the NetCDF4 library.

The following magnetic field components were extracted:

- B_x
- B_y
- B_z

The magnetic field magnitude was calculated using

$$B = \sqrt{B_x^2 + B_y^2 + B_z^2}$$

The average magnetic field magnitude was then computed.

8. Results

The detected active region intensity and magnetic field measurements are summarized below.

Parameter	Value
Mean Active Region Intensity	0.665
Mean Magnetic Field Magnitude	303.72 nT
Pearson Correlation	Not Defined (NaN)

9. Intensity–Magnetic Field Comparison

The comparison between image intensity and magnetic field magnitude is illustrated in Figure 2.

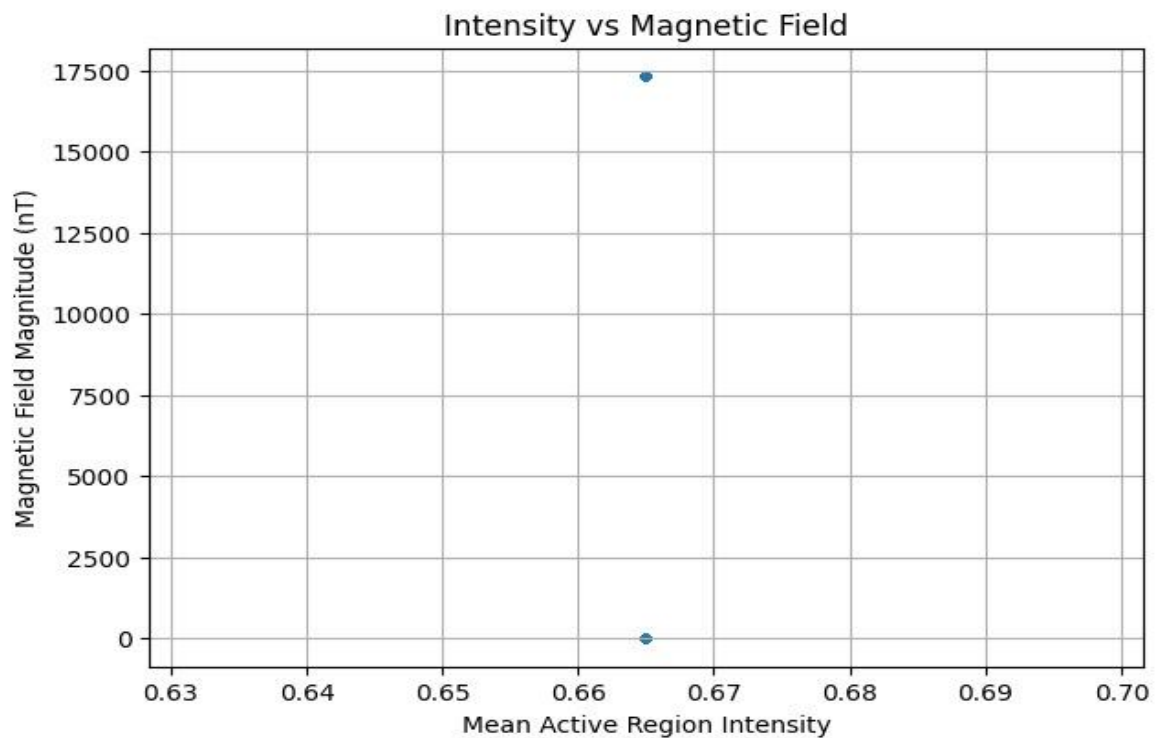


Figure 2. Intensity vs Magnetic Field

The graph shows the measured average intensity of the detected active region together with the corresponding magnetic field measurement obtained from the MAG instrument.

10. Discussion

The active region was successfully detected using image segmentation.

The average intensity of the detected region was found to be approximately **0.665**.

The average magnetic field magnitude calculated from the MAG dataset was approximately **303.72 nT**.

The Pearson correlation coefficient could not be computed because the available MAG dataset provides spacecraft magnetic field measurements instead of pixel-wise magnetic field images.

Since only one representative magnetic field value corresponds to the observation, sufficient statistical variation required for Pearson correlation is unavailable.

Therefore, only a qualitative comparison between intensity and magnetic field measurements could be performed.

11. Limitations

The following limitations were observed during this study.

- The MAG dataset does not contain image-based magnetograms.
- Pixel-to-pixel comparison is not possible.
- Pearson correlation requires multiple paired observations.
- Only one observation was available for comparison.

12. Future Work

Future work may include:

- Using multiple SUT observations.
- Downloading HMI/SDO magnetogram images.
- Performing pixel-wise intensity–magnetic field correlation.
- Applying Pearson and Spearman correlation analysis.

- Investigating temporal evolution of active regions.

13. Conclusion

This work successfully demonstrated the detection of solar active regions from Aditya-L1 SUIF images and the extraction of magnetic field measurements from the Aditya-L1 MAG dataset.

The detected active region exhibited a mean brightness intensity of **0.665**, while the average magnetic field magnitude measured by the MAG instrument was approximately **303.72 nT**.

Although a direct statistical correlation could not be established due to the absence of pixel-wise magnetogram data, the implemented methodology provides a foundation for future image–magnetic field correlation studies using spatial magnetogram datasets.

References

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PART 2C

Comparative Segmentation Methods for Solar Image Analysis Using Aditya-L1 SUIT Data

1. Introduction

Solar image segmentation is an important step in identifying active regions and studying solar activity. Different image processing techniques can be used to separate bright solar structures from the background. In this work, Otsu Thresholding, Adaptive Thresholding, and K-Means Clustering were applied to an Aditya-L1 SUIT solar image and their performance was compared.

2. Objective

The objective of this study is to compare different image segmentation methods and evaluate their effectiveness in identifying solar active regions from Aditya-L1 SUIT images.

3. Dataset Information

The dataset used in this study was obtained from the Solar Ultraviolet Imaging Telescope (SUIT) onboard India's Aditya-L1 mission. SUIT captures high-resolution images of the Sun in ultraviolet wavelengths and provides valuable information about solar activity and surface features.

The image used for analysis was stored in FITS (Flexible Image Transport System) format, which is widely used in astronomy for scientific data storage.

Dataset Details

- Mission: Aditya-L1
- Instrument: Solar Ultraviolet Imaging Telescope (SUIT)
- Data Format: FITS (.fits)
- Filter: NB03
- Observation Date: 24 June 2026

The dataset was used to perform image segmentation and compare different techniques for identifying solar active regions.

4. Methodology

The FITS image was loaded using Astropy and processed using Python-based image analysis techniques. Invalid values were removed and the image was prepared for segmentation. Three segmentation methods were then applied:

- Otsu Thresholding
- Adaptive Thresholding
- K-Means Clustering

The segmentation results were compared based on their ability to detect bright solar structures and active regions.

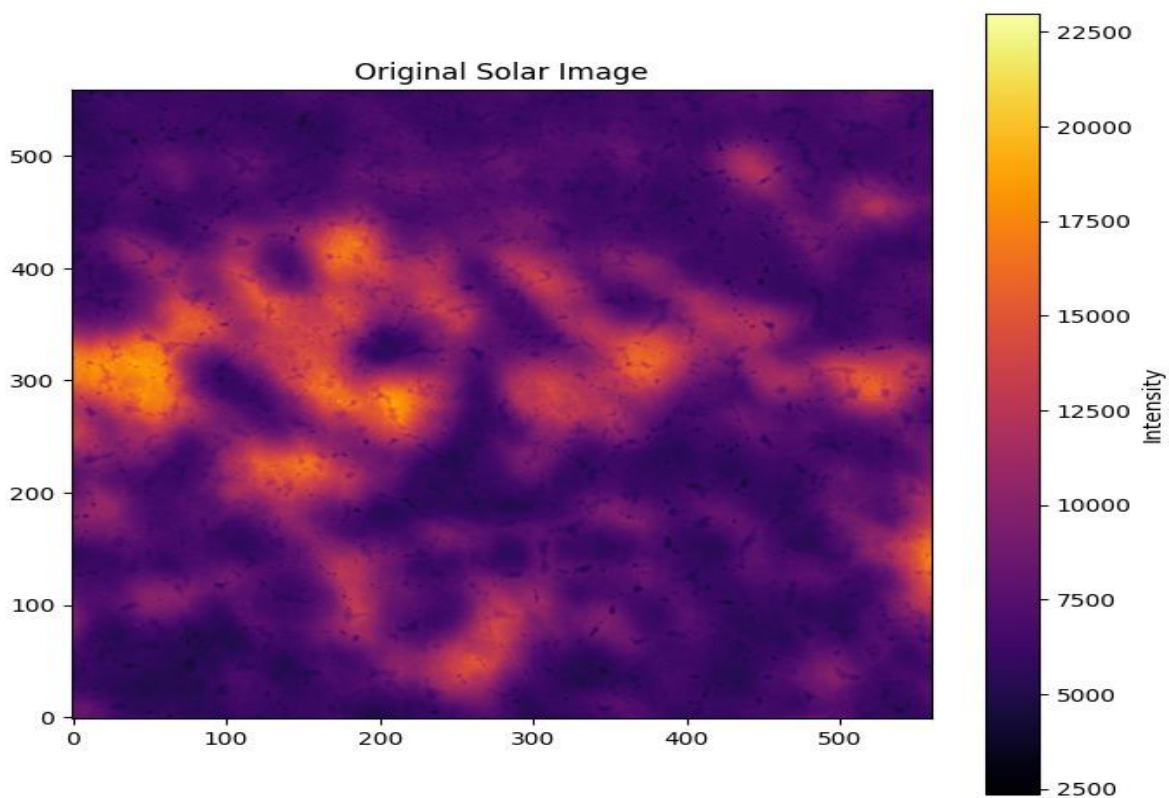


Figure 1. Original Solar Image

5. Flowchart

START



Load FITS Image



Image Preprocessing



Apply Otsu Thresholding



Apply Adaptive Thresholding



Apply K-Means Clustering



Compare Results



Generate Graph



END

6. Algorithm

Algorithm: Comparative Segmentation Methods

1. Load the SUIT FITS image.
2. Extract image intensity data.
3. Remove invalid pixel values.
4. Apply Otsu Thresholding.
5. Apply Adaptive Thresholding.
6. Apply K-Means Clustering.
7. Count detected pixels for each method.
8. Generate comparison graph.
9. Compare segmentation performance.
10. Display and save results.

7. Otsu Thresholding

Otsu Thresholding automatically determines an optimal threshold value and separates bright solar regions from the background. This method produces a clean binary segmentation and effectively identifies major bright regions.

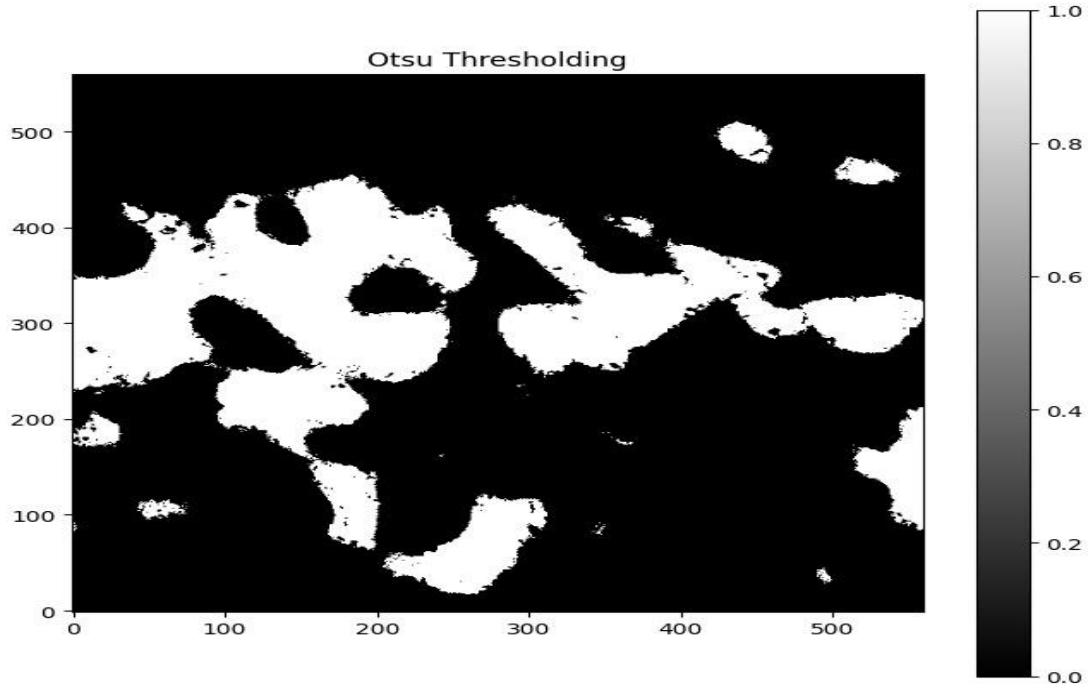


Figure 3. Otsu Thresholding Result

8. Adaptive Thresholding

Adaptive Thresholding calculates local threshold values for different image regions. This method is capable of detecting smaller structures and local intensity variations but may introduce additional noise.

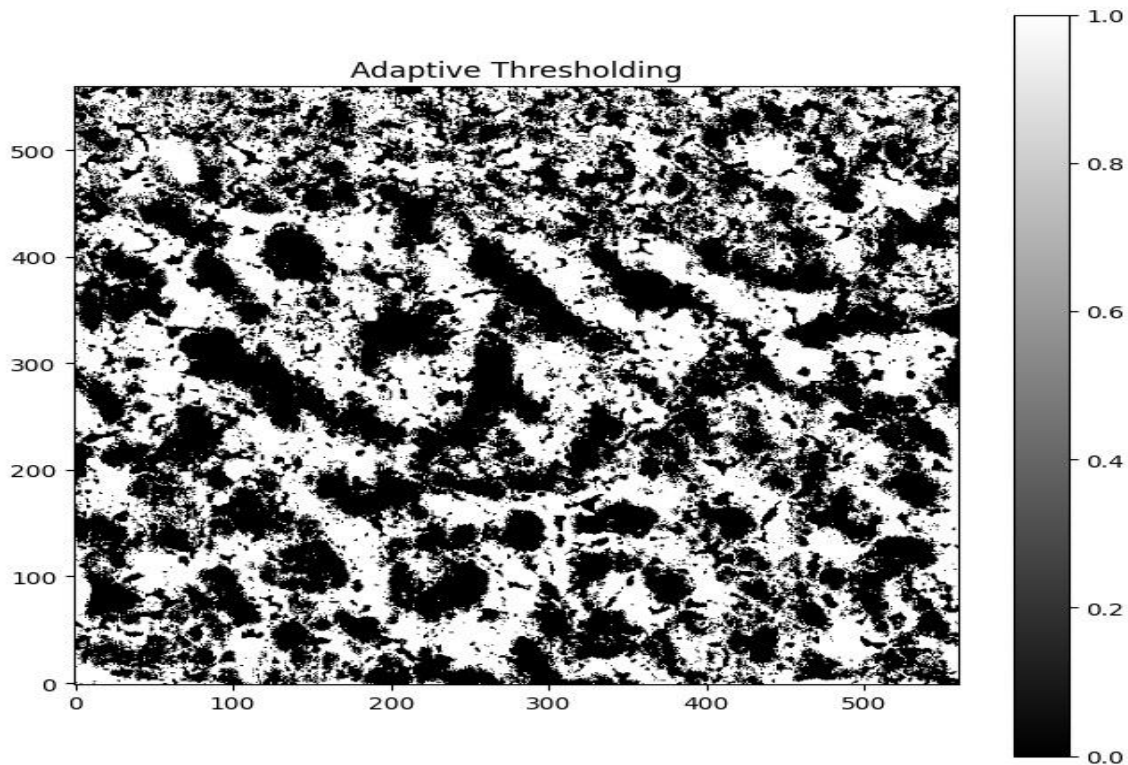


Figure 4. Adaptive Thresholding Result

9. K-Means Clustering

K-Means Clustering groups pixels according to their intensity values and performs segmentation without requiring a predefined threshold. This method separates the image into intensity-based regions.

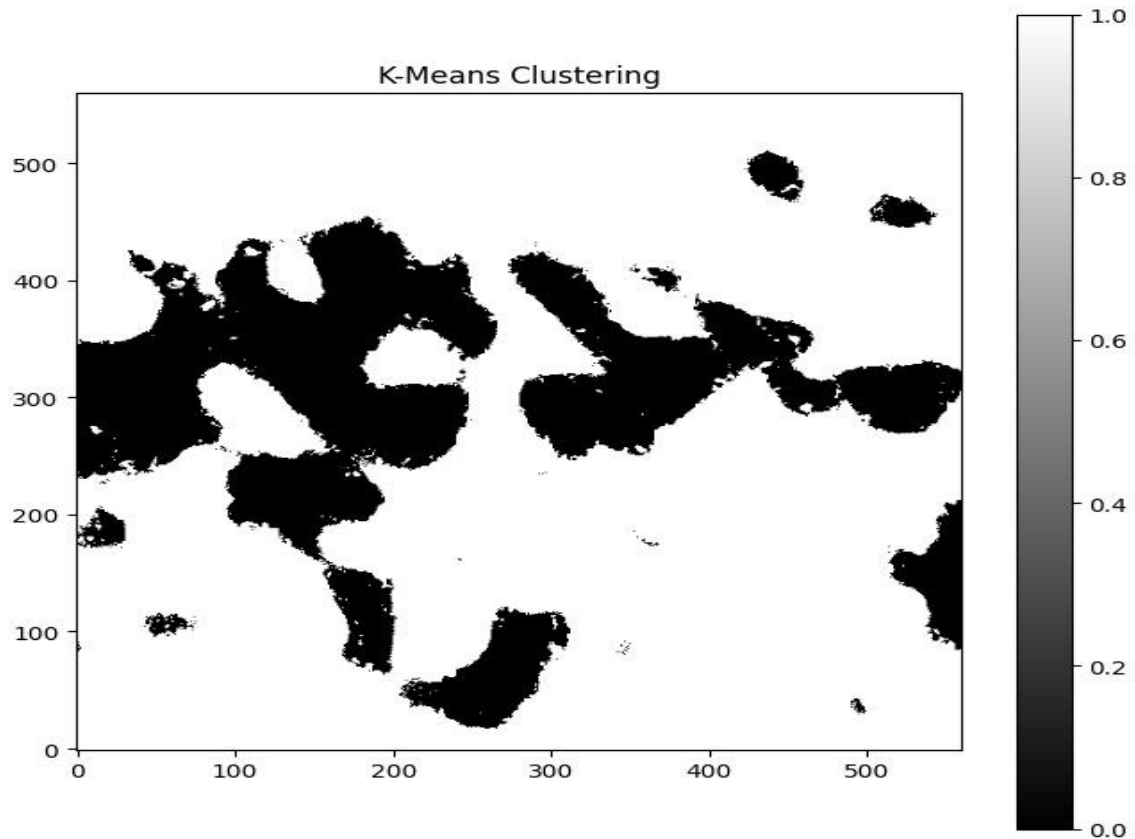


Figure 5. K-Means Clustering Result

10. Results and Discussion

The solar image was successfully segmented using Otsu Thresholding, Adaptive Thresholding, and K-Means Clustering.

Otsu Thresholding produced a clean segmentation result with minimal noise and clearly identified the major bright regions present in the solar image.

Adaptive Thresholding detected a larger number of local structures because threshold values were calculated for smaller image neighborhoods. However, the method also introduced additional noise and fragmented regions.

K-Means Clustering grouped pixels according to their intensity values and detected the largest number of pixels among the three methods.

The comparison graph shows that Otsu Thresholding detected approximately 72,000 pixels, Adaptive Thresholding detected approximately 151,000 pixels, and K-Means Clustering detected approximately 242,000 pixels.

The results indicate that Otsu Thresholding provides the cleanest segmentation, Adaptive Thresholding captures more local details, and K-Means Clustering provides effective intensity-based classification.

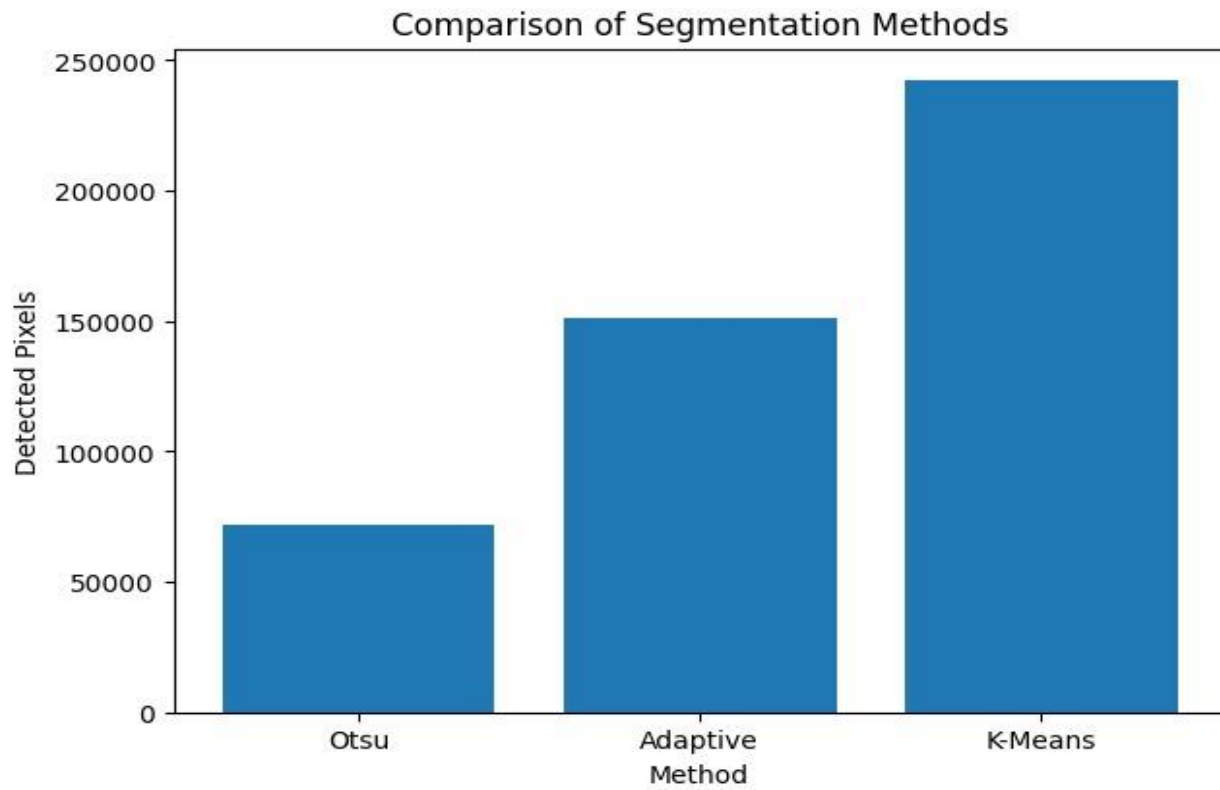


Figure 6. Comparison of Segmentation Methods

11. Conclusion

In this study, three image segmentation methods, namely Otsu Thresholding, Adaptive Thresholding, and K-Means Clustering, were applied to an Aditya-L1 SUIT solar image.

The results showed that Otsu Thresholding produced a clean and noise-free segmentation of major bright regions. Adaptive Thresholding detected more local features but introduced additional noise. K-Means Clustering successfully

classified the image into intensity-based regions and detected the largest number of pixels.

Overall, Otsu Thresholding was found to be the most suitable method for identifying prominent active regions in the selected solar image. The study demonstrates the usefulness of image processing techniques for automated solar image analysis using Aditya-L1 data.

12. References

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